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GZX Series (300℃)

DRYING OVEN

Instruction Manual <<<

Shanghai Boxun Medical Biological Instrument Corp .

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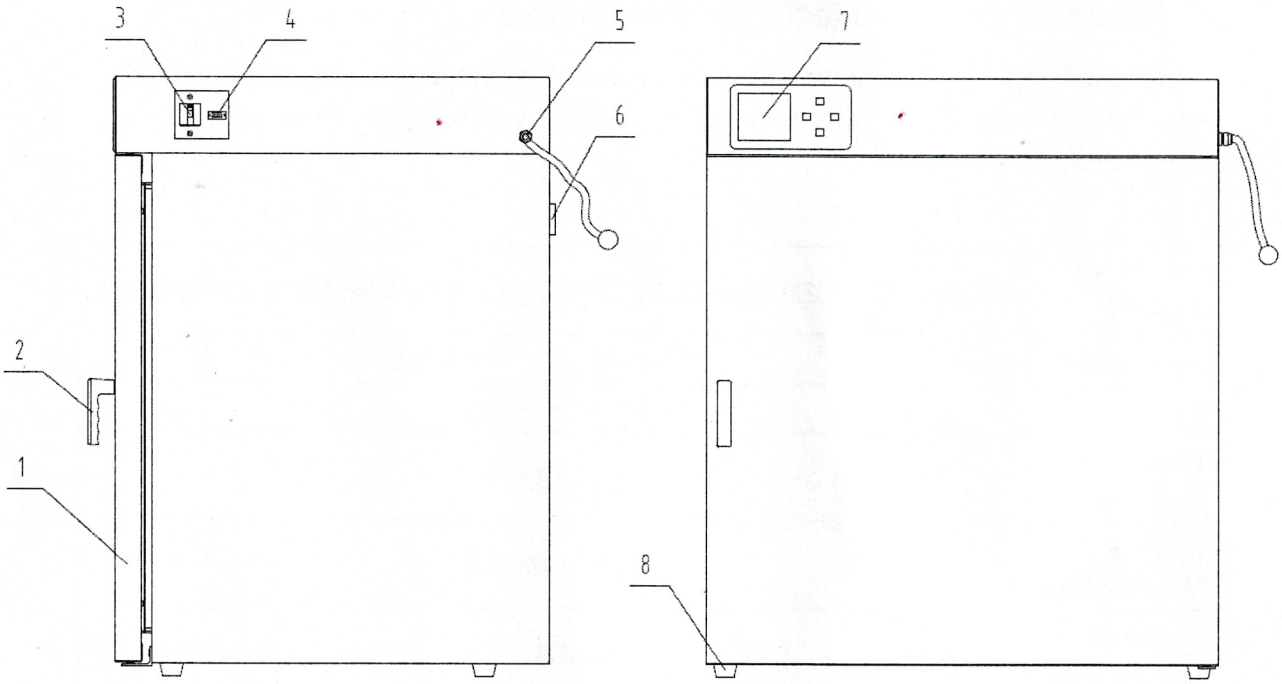
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Thank you very much for your purchasing products from Shanghai Boxun Medical Biological Instrument Corp. To allow the equipment maximizing the effectiveness for you, please take the originals or the copies of the product manual to users. You must be familiar with the details of this manual, and it will let you master the essentials in the course of operation and make your work smoothly. Please keep this manual safety so that you can check at any time.

I. Diagram of product structure



1、Door　　2、Handle　　3、Power Switch

4、485 interface　　5、Power cord　　6、Exhaust Hole

7、Operation panel　　8、Feet

II. Product overview

Drying ovens are widely used in biochemistry, chemical pharmaceutical, industrial and mining enterprises, colleges and universities, scientific research and other fields for drying, baking, wax melting, disinfection and sterilization.

The shell of drying oven is made of high-quality cold rolled steel plate, the surface is sprayed, the studio is made of high-quality stainless-steel material, and the heat insulation layer is made of centrifugal glass fiber material. The heat-resistant silicone rubber sealing ring is installed between the studio and the outer door, which greatly increases the sealing ability of the door and the studio, and greatly improves the insulation performance of the oven body. The drying oven adopts intelligent temperature control system, which has accurate temperature control, high precision and timing function.

III. Product model and technical parameters

Please confirm carefully that whether the model of equipment is consistent with the model you chosen. If any difference, please contact with dealers or our company in time to exchange it.

Model / Parameter	GZX-9076MBE	GZX-9146MBE	GZX-9246MBE	GZX-9306MBE	GZX-9456MBE
Power supply	AC 220V±10V 50/60HZ				
Temp. range	RT+5℃～300℃				
Temperature uniformity	±2.5%（150℃）				
Temp. fluctuation	±1℃（150℃）				
Power consumption	1550W	2150W	2750W	2750W	3500W
Internal Dimensions(mm) W×D×H	410×350×450	570×400×570	630×530×630	720×530×745	600×580×1300
Housing Dimensions(mm) W×D×H	555×566×722	715×616×842	775×746×902	865×746×1017	774×845×1704
Volume (L)	64	129	210	284	452
Optional features	Printer (support curve), GPRS remote control, computer monitoring three options				

Note: All data are only applicable to non-loaded equipment with ambient temperature of 25±3℃ and power supply voltage of 220V±10%. Data is the average value of series equipment, the

company reserves the right to change.

IV. Installation method and requirements

> ⚠ Warning
>
> This equipment should not be used outdoors. If the equipment is happening to be in water or under the rain, electric shock may occur.
>
> This equipment should not be installed in wet locations or where there may be sources of accidental liquid spills. Water will reduce the degree of insulation during an electric shock.

4.1 Installation Environment

In order to make the normal operation of the equipment and get the best performance, please install the equipment at the place comply with the following conditions:

■ Ambient Temperature: 5～35 ℃; Relative Humidity＜80%; Atmospheric Pressure: 86Kpa～106Kpa.

■ A place that is not exposed to direct sunlight and has no strong vibrations and airflow around it.

Do not install the equipment in direct sunlight, strong air flow and vibration, or it may cause the equipment to fail to achieve the expected performance.

■ A place where there is plenty of space.

The equipment must be above 30cm distance from the wall.The distance between the two sides is more than 15cm.

■ A place far from the heat source.

Avoid installing equipment near heat sources such as boilers or heaters. Excess heat from the outside will affect the expected performance of the equipment.

■ A place with a solid and smooth surface.

Be sure to install equipment on solid and flat ground, uneven ground or tilted installation may cause malfunction or injury to personnel. Install the equipment in stable conditions to avoid sloshing and noise.

■ A place where high humidity is not likely to occur.

Install the equipment at locations where humidity is less than 80%. If installed in a high humidity location, it may cause leakage or electric shock.

■ A place where there is no flammable or corrosive gas.

Do not install equipment where flammable or volatile substances are present or may cause explosion or fire. Or it may cause leakage, electric shock or damage to equipment as a result of corrosion.

4.2 Installation Method

4.2.1 Take off the packing material

After removing the packing box, open the box door, take out the random accessories and technical documents, and check the packing list.

4.2.2 Installation of oven

(a) Power installation: there should be a reliable power source near the equipment, the power supply capacity should meet the requirements of the equipment, and fuses or air circuit breakers should be installed in the power supply circuit to ensure the safety of power supply and then connect the power supply.

b) plug-connected devices (AC 220 V) can be directly plugged into power sockets. After the power connection, the installation of the equipment has been completed and can be put into use.

4.3 Pre-operation preparation

Before starting the device for the first time, follow the following procedures:

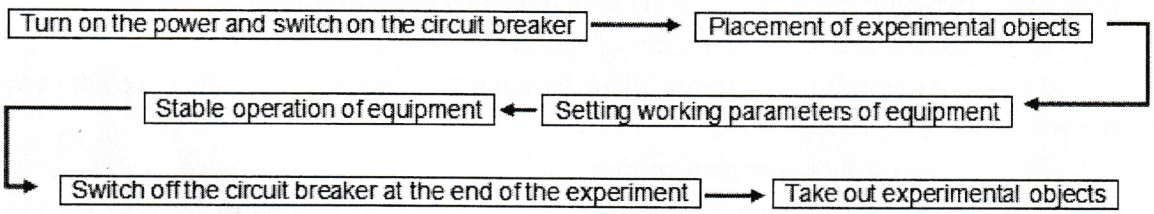
a) removing shelves, etc., from the inside of the equipment.

b) sterilize the inner wall of the chamber with soaked gauze, then wipe the alcohol off with dry gauze.

c) placing the shelves into the equipment in accordance with experimental spacing requirements.

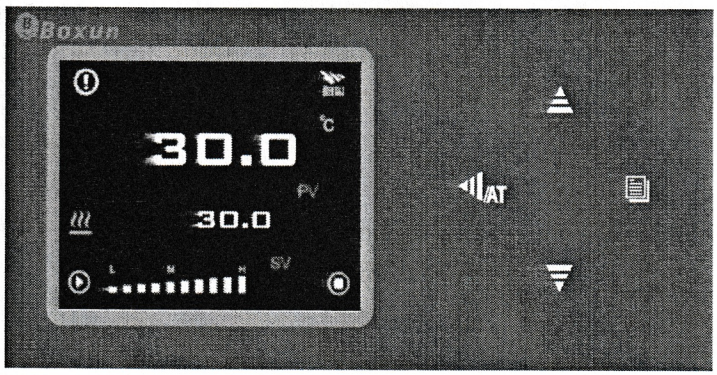
V. Operation instruction

5.1 Flow of work



5.2 Home screen and key instructions

(1)  (Setting button) : Press once to enter the setting of temperature and time; Long press to enter internal parameter setting.



(2)  Shift/self-setting key : Press this key for 6 seconds in non-setting state to enter or exit system self-setting.

(3)  Reduce/Re-run key : in the non-set state, when the operation is finished, long press this key for 3 seconds to restart the operation. In multi-value mode, click to display the current operation cycle, and then click to display the number of current operation segments. In the set state, click this key to

make the set value decrease, long press this key to make the set value decrease continuously.

(4)  Add key : Click this key to increase the set value in the set state, long press this key to continuously increase the set value, click this key to turn on or off the LCD backlight in the non-set state.

(5) Display Identifier meaning

PV ———————— Inner temperature display

SV ———————— Set temperature display

 ———————— This light is on when running and off when ending

 ———————— This light is off at run time and on at end

 ———————— This lamp is on if there is heating , otherwise it is off

 ———————— This light will be on when overtemperature alarm is reported

5.3 Use method (Steps)

1) Please put the items that need experimentalized on the chamber shelf, and close the door.

Note:

The notes about placing the training materials in the Drying Oven: Please put the items that need dried or experimentalized on the shelf sequentially, put the items on one shelf, and elapse space for air convection. otherwise, our company won't handle the fault. Thank you for your cooperation.

The notes about opening the door with handle: The door is opened with the handle, and the handle is opened vertically, raise the handle slowly along the vertical direction by the arc to break away the agraffe, and the door can be opened at this time; press the handle slowly along the vertical direction by the arc to close the agraffe, and the door can be closed at this time.

2) Please confirm that your equipment's power supply has accepted to 220V power supply outlet and the power light on the panel is on; press the switch button "O/I" to "I", and the power switch light is on at this time, it shows that the power has been sent

to the equipment.

3) After the power supply is turned on, the PV/SV display window shows the sensor type and the maximum temperature measurement. After a few seconds, the PV display window displays the measured value of the current temperature, and the SV display window displays the set value of the temperature. At this time, the equipment works according to the set temperature.

4) In the display state, you can set the temperature and timing time required by your experimental work and other functions, specific operations are as follows:

A. Temperature setting: Click the "Set" key to enter the temperature setting state, and the word "SP-" is displayed in the SV display window. Set the required temperature value in the PV display window by the shift key, decrease key and increase key.

B, time setting, press "set" key twice, the SV display window "St-", enter the time setting, through the shift key, reduce and increase the keys in the PV display window to set the desired value, and then press the "set" button, return to the temperature display interface (timing time setting unit: minutes), the modified value is automatically saved.

C. Timing function description: When the time is set to "0", it indicates that there is no timing function and the controller runs in a continuous cycle. When set to "0", such as measuring temperature reach the set temperature, the timer start time, stop running after the timing, the SV display window "End", at the same time a buzzer sing 30 seconds (when the buzzer sounds, press any key censored), long press "reducing/to run" key 3 seconds, the program can start running again.

Note

Special note about timing function: this drying box has timing function, the range of timing is 1-9999 minutes, the minimum set time unit is 1 minute. When the set time is 0 minutes, the device is in continuous working state. After the device is started, it works according to the previous set timing program. If you do not need to use the timing function, please set the timing time to 0.

5) Self-setting function description: When the temperature control effect is not ideal, the system can be self-setting. In the process of self-tuning, there will be a large overshoot in the temperature. Users should fully consider this factor before carrying out system self-tuning. Before self-tuning, keep the temperature in the box at room temperature.

Operation method: Press the "Shift/Self-setting" button for 6 seconds and enter the system self-setting program. The indicator blinks. After the self-setting, the indicator blinks and the controller gets a set of better PID parameters, and the parameter values are saved automatically. When the self-setting is finished, it automatically enters the normal working state. The instrument does not respond to any operation during the self-setting process. Please wait patiently. If you need to stop the self-setting, press the "Shift/Self-setting" button for 6 seconds during the system self-setting process to stop the self-setting process or turn off the power.

In the process of system self-tuning, if there is an over temperature alarm, "!" The alarm light does not light, nor does the buzzer, but the heating alarm relay will automatically disconnect. The Set key is invalid during system self-tuning. In the process of system self-tuning, regardless of whether there is a constant temperature time setting, the lower row of the controller display window always displays the temperature setting value.

Note:

Special notes on the self-tuning function:

A. Do not cut off the power supply or open the door during the self-setting process. If the self-tuning abnormal interrupt, the process is invalid, and the original parameters will not change.

B. When you start the machine for the first time or the machine works in an environment where the ambient temperature changes greatly (more than 20°C), you must start the self-setting function.

C. In the process of equipment self-tuning, it is necessary to maintain the continuity of power supply and minimize the interference of power grid to the equipment; otherwise, self-tuning needs to be restarted.

D. Start the self-setting function after startup, and find that some fluctuations cannot meet your requirements after constant temperature, so you can choose to turn on the self-setting function when the temperature is close to your requirements, which may meet your requirements.

E. In the case of great interference, the method of multiple self-setting can be adopted to check the rationality of the parameters after setting and determine them. If you choose to amend, you can inquire with the unit separately.

F. When the required working temperature is low, the second setting mode can be adopted. If the required working temperature is 60°C, the first setting can be set at 50°C.

6) When experimentalizing ends, please cut off the power supply. After the items cool to a certain temperature (room temperature is best), open the door and take out the items. Please be careful of the temperature and don't be burned.

Please operate the equipment as above steps for the next time. Thank you.

Notes of the use of equipment:

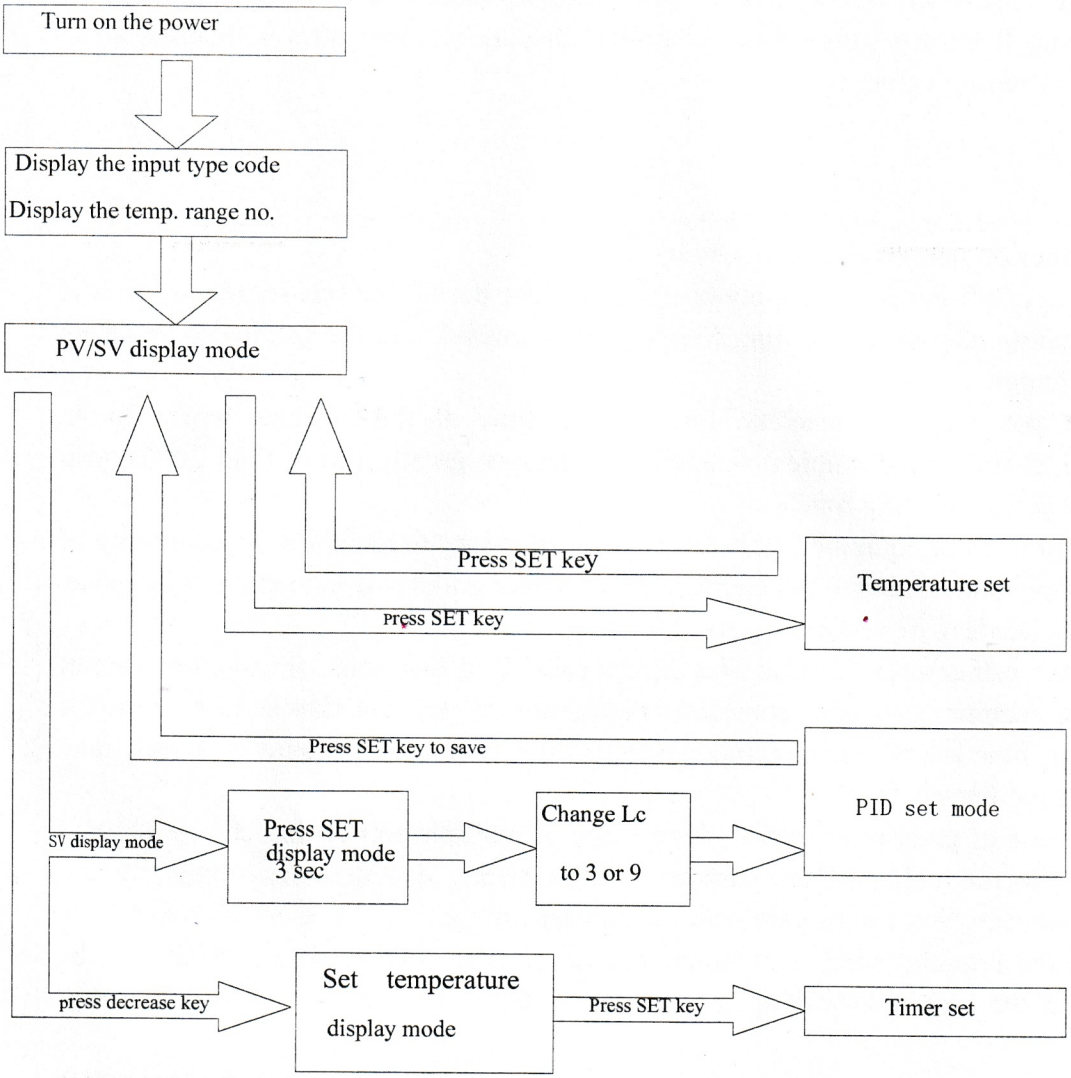
1. Drying Oven adopts self-adaption temperature control, and auto-turning is unnecessary.

2. The Drying Oven should have reliable grounding to ensure the safe use.

3. If the items are flammable, you must take them out when temperature is lower than the burning point to avoid oxidation reaction and burn.

4. No explosion-proof device, don't take the explosive items into the Drying Oven for training.

5.4 Control panel setting steps



VI.The parameters on the control panel indicate meanings

Long press "set" key about 3 seconds, a password prompt "Lc -", by reducing the key and key password, click on the "set" button, to the interior temperature parameter set state, click the set key, reduce and increase the keys can modify the parameters, press "set" key for 3 seconds longer to exit the state, and automatically save the set parameters. If no key is pressed within 30 seconds, the system automatically exits this state and the current parameters are not saved.

Symbol	Symbol	Symbol	Symbol
Lc-	Code	When Lc=3, parameter value can be seen and amended	0
AL-	Over-temp. alarm	When "temperature measurement value > When the temperature set value +AL ", the alarm light is on, the buzzer buzzes, and the heating output is disconnected	(0～100℃) 5.0
T-	Control period	Control period	(1～60S) 1
P1-	Ratio 1	Time ratio adjustment 1, only temperature SP< Dc effective	(1.0～Range value)35.0
I1-	Integration time 1	Integral action regulation 1, only temperature SP<Dc is effective.	(1～1000s) 200
d1-	Differential time 1	Adjust differential effect , only temperature SP<Dc is effective.	(0～1000s) 200
P2-	Ratio 2	Time ratio adjustment 1, only temperature SP≥Dc effective	(1. 0～Range value) 35. 0
I2-	Integration time 2	Adjust differential effect,only temperature SP≥Dc is effective.	(1～1000s) 200
d2-	Differential time 2	Adjust differential effect , only temperature SP≥Dc is effective.	(0～1000s) 200
dc-	Inflectiontemperature	When the set temperature ≤ DC, belongs to the low temperature zone	(0～Range value) 80
Pb-	Zero adjustment	Correction of sensor (low temperature) measurement errors. Pb= actual temperature value - instrument measurement value	(-50～50℃) 0
PK-	Full scale adjustment	Amend the error brought by sensor (high temperature) **PK** =1000*(actual temp. value – measured temp. value) / measured temp. value	(-999～999) 0
Et-	Timing function	When ET=0, there is no timing function. When it is 1, the timer starts to be powered on; when it is 2, the timer starts to be set	(0～2) 2

Note: In order to achieve the best effect of the control, P can be appropriately modified (the modification range is generally between "30 and 40"), but usually there

is no need to modify, directly use the factory parameters.

Temperature correction method:

1. When Lc is displayed in the upper display window, press the Decrease key and the Increase key and enter "3" in the lower display window.

2. Press the "Set" key again to enter the internal parameter setting mode.

3. Adjust the " Pb- "and" PK- ", and press "Set" to return after the correction.

Judgment premise:

1. The thermometer shall be measured after passing the inspection of the measuring unit.

2, mercury thermometer measurement installation position: please hang the thermometer in the center of the cavity, can not be directly placed on the partition board measurement.

VII.Maintenance and repair

In daily use, although the operation has been carried out as required, there will still be unexpected failures. In this case, please make self-diagnosis according to the "Fault cause check table", and handle some faults that you can troubleshoot by yourself. For some failures that customers cannot solve, please record the equipment model specification, product number and contact with the local dealer or our company to obtain maintenance services.

Equipment malfunctions, analysis and solutions:

Fault phenomenon	Fault diagnosis	Handle
Sensor fault alarm	• Temperature sensor is abnormal, check temperature sensor, model PT100.	• Contact supplier or manufacturer
The temperature does not reach the set value	• Check that the electric heating pipe is heated. • Check the silicon control. • Check if the thermal switch is on.	• Contact supplier or manufacturer • Exchange the silicon control
Display screen without display	• Check whether circuit breaker is damaged. • Check that the power switch is energized. • Check if 220V power supply on the incoming plug line.	• Exchange the circuit breaker • Turn on the power switch • Restoring power supply

After-sales service information

a） The warranty is one year from the product is purchased. After that, according to the equipment failure, charge for service and maintenance of spare parts as appropriate.

b） Our company does not undertake the obligation for the damage caused by without using in accordance with the specification requirements or disassemble and refit of the equipment without authorization.

c） If users need maintenance services, technical advice, accessories supply, please contact us, we will be dedicated to your service.

> ⚠ Caution:
> Power must be cut off during maintenance.

The oven is a high-precision constant temperature experimental equipment. Daily maintenance can effectively improve the service life of the equipment and ensure the effectiveness of its use. Therefore, the following matters should be noted in the daily operation:

■ Power connections should be checked daily and should be safe and reliable.

■ Keep equipment and environment clean.

■ The inner wall of the chamber should be cleaned regularly. The shelf should be removed before cleaning. It can be wiped with neutral detergent or alcohol.

■ When moving the equipment, please fix the accessories in the chamber.

■ Check and confirm the equipment test parameters before each use to ensure the safe operation of the equipment.

■ In order to ensure the sealing of the chamber, the integrity and cleanliness of the sealing ring of the door should be checked and maintained frequently, and the broken sealing ring should be replaced in time.

VIII.Packing list

No.	Name	Qty	Remarks
1	Equipment	1 pc	
2	Instruction Manual	1 pc	
3	Certificate	1 pc	
4	Quality and Warranty Card	1 pc	
5	Shelf	GZX-9076MBE 2 pcs	
		GZX-9146MBE 2 pcs	
		GZX-9246MBE 3 pcs	
		GZX-9306MBE 3 pcs	
		GZX-9456MBE 4 pcs	